

ASSIGNMENT 2.1

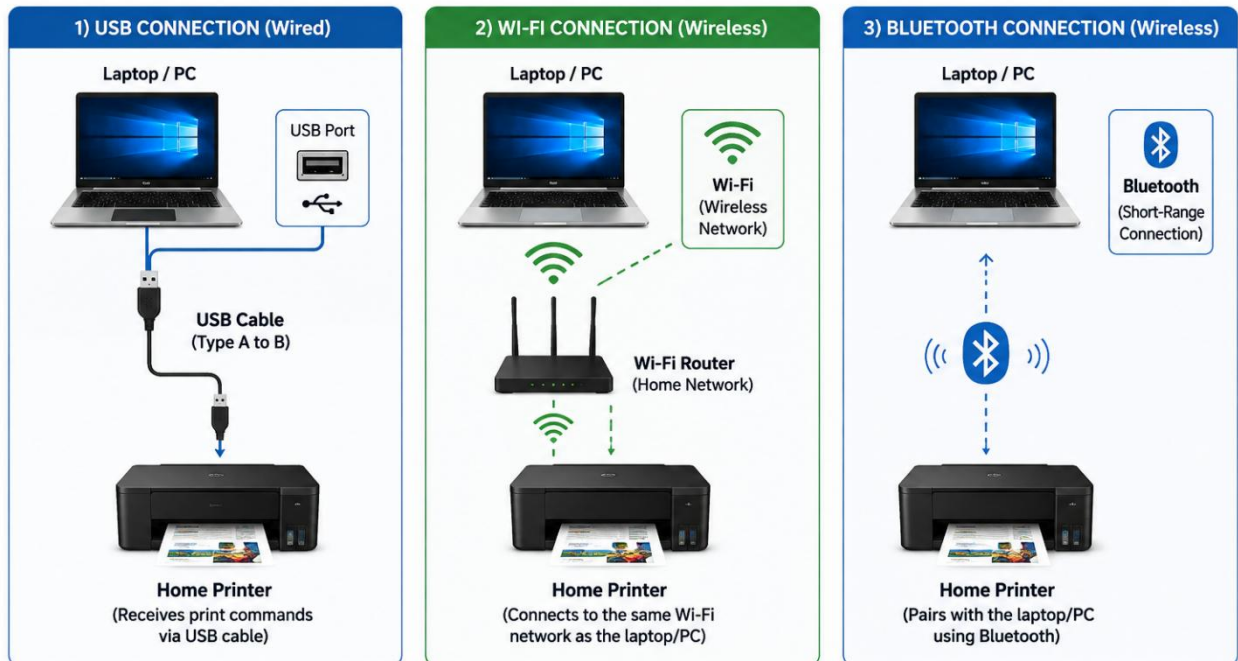
Q1. Create a table listing 3 types of printers, 2 types of scanners, and 2 types of webcams, and for each, write one real-life use case (for example: 'Inkjet printer – printing college assignments').

Devices	Type	Uses
Printer	Inkjet Printer	Printing photo-quality family pictures and college assignments at home.
Printer	Laser Printer	Rapidly printing hundreds of text-heavy contracts and invoices in an office.
Printer	Thermal Printer	Printing instant paper receipts at a grocery store checkout register
Scanner	Flatbed Scanner	Digitizing fragile old family photos, book pages, or official paper documents.
Scanner	Barcode / QR Scanner	Scanning items at a retail store checkout or tracking packages in a warehouse.
Webcam	Integrated / Built-in Webcam	Attending quick remote work meetings or online classes via a laptop.
Webcam	External USB Webcam	Streaming games on Twitch or hosting professional webinars in 1080p/4K resolution.

Q2. Draw a labelled diagram showing how a typical home printer connects to a laptop or PC, including the connection interface used (USB, Wi-Fi, or Bluetooth).

How a Home Printer Connects to a Laptop / PC

Connection Interfaces: **USB** | **Wi-Fi** | **Bluetooth**



Note: Make sure the printer is installed with proper drivers/software on the laptop/PC before printing.

Q3. List three different connection interfaces (such as USB, Wi-Fi, Bluetooth) used by peripherals like printers, scanners, and webcams, and for each, name one advantage and one disadvantage. Hint: Think about speed, convenience, and compatibility.

1. USB (Wired)

Advantage: Highly reliable with fast data transfer speeds and zero signal latency or interference.

Disadvantage: Physical cable clutter limits mobility and restricts placement distance from the host computer.

2. Wi-Fi (Wireless Network)

Advantage: Enables multi-device sharing across a local network without needing direct physical cables.

Disadvantage: Network setup can be complex, and connection quality relies heavily on router stability and signal strength.

3. Bluetooth (Direct Wireless)

Advantage: Simple, low-power short-range connection that pairs directly without requiring a local Wi-Fi router.

Disadvantage: Slower data transfer rates compared to USB or Wi-Fi, making it less ideal for high-bandwidth tasks like high-resolution video streaming.

Q4. Pick any one device you use at home or college (printer, scanner, or webcam). Write step-by-step instructions for how you would connect it to your computer, including what cable or wireless method you would use and any setup steps required.

Required Items:

- USB Webcam
- Laptop or Desktop PC
- Internet connection (for automatic driver installation)

Step-by-Step Setup Instructions:

1. **Mount the Webcam:** Open the adjustable clip at the base of the webcam and place it securely on top of your monitor or laptop screen. Angle the lens so it faces your face directly.
2. **Plug in the USB Cable:** Insert the webcam's USB cable into any available **USB Port** on your computer or laptop.
3. **Driver Installation (Automatic):** Modern operating systems (Windows and macOS) automatically detect the device. A notification like *"Setting up device..."* will appear as the system installs the required drivers. Wait 1–2 minutes for this to finish.
4. **Test the Webcam:**
 - **On Windows:** Open the **Camera** app from the Start menu.
 - **On Mac:** Open **Photo Booth** or **FaceTime**.
 - If your video feed appears on screen, the webcam is working properly.
5. **Select in Meeting Apps:** Open your preferred video calling app (such as Zoom, Google Meet, or Microsoft Teams), go to **Settings > Video**, and choose your new USB Webcam as the primary video device.